Fundamentals of Electrical Engineering: Star-Delta Transformation

1. High-Level Overview

In electrical circuit analysis, we frequently encounter complex resistor networks that cannot be solved using simple **Series** (resistors in a single line) or **Parallel** (resistors connected across the same two points) rules. When resistors are arranged in interconnected loops, network reduction becomes difficult.

To simplify these complex networks, we use **Star-Delta ($\Upsilon - \Delta$) Transformations**. This technique lets us replace a three-terminal sub-network connected in a **Star** (or "Y") configuration with an equivalent three-terminal network connected in a **Delta** (or "Δ") configuration, and vice versa.

- **Equivalence Principle:** The transformation is mathematically constructed such that the electrical behavior (voltage drops, line currents, and overall power consumption) measured at the external terminals A, B, and C remains completely identical before and after conversion.

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2. Structural Configurations & Conversion Formulas

Diagram: Star and Delta Resistor Networks

Description of Diagram: On the left is a Star (Y) network with three terminals labeled A, B, and C. Three resistors (R_A, R_B, R_C) meet at a central common node (neutral point). R_A connects to terminal A, R_B to terminal B, and R_C to terminal C. On the right is a Delta (Δ) network forming a triangle with vertices labeled A, B, and C. Resistor R_{AB} connects terminals A and B, resistor R_{BC} connects terminals B and C, and resistor R_{CA} connects terminals C and A. A double-headed arrow between both networks indicates mutual equivalence.

A. Delta-to-Star Conversion ($\Delta \to \Upsilon$)

Goal: You have a closed triangular Delta network (R_{AB}, R_{BC}, R_{CA}) and want to convert it into a three-armed Star network (R_A, R_B, R_C).

- **Intuitive Memory Rule:** Any Star arm resistor connected to a terminal equals the **product of the two adjacent Delta resistors** connected to that same terminal, divided by the **sum of all three Delta resistors**.

Let $\sum R_\Delta = R_{AB} + R_{BC} + R_{CA}$ represent the sum of all Delta resistors.

1. **Resistor attached to Terminal A:**

$$R_A = \frac{R_{AB} \cdot R_{CA}}{R_{AB} + R_{BC} + R_{CA}} = \frac{R_{AB} \cdot R_{CA}}{\sum R_\Delta}$$

2. **Resistor attached to Terminal B:**

$$R_B = \frac{R_{AB} \cdot R_{BC}}{R_{AB} + R_{BC} + R_{CA}} = \frac{R_{AB} \cdot R_{BC}}{\sum R_\Delta}$$

3. **Resistor attached to Terminal C:**

$$R_C = \frac{R_{BC} \cdot R_{CA}}{R_{AB} + R_{BC} + R_{CA}} = \frac{R_{BC} \cdot R_{CA}}{\sum R_\Delta}$$

B. Star-to-Delta Conversion ($\Upsilon \to \Delta$)

Goal: You have a central Star network (R_A, R_B, R_C) and want to convert it into an outer triangular Delta network (R_{AB}, R_{BC}, R_{CA}).

- **Intuitive Memory Rule:** Any Delta resistor connected between two terminals equals the **sum of the two adjacent Star resistors**, plus the **product of those two Star resistors divided by the third (opposite) Star resistor**.

Let $\sum(R_{\text{pair}}) = R_A R_B + R_B R_C + R_C R_A$ represent the sum of pairwise products of Star resistors.

1. **Resistor between Terminals A and B:**

$$R_{AB} = R_A + R_B + \frac{R_A \cdot R_B}{R_C} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_C} = \frac{\sum(R_{\text{pair}})}{R_C}$$

2. **Resistor between Terminals B and C:**

$$R_{BC} = R_B + R_C + \frac{R_B \cdot R_C}{R_A} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_A} = \frac{\sum(R_{\text{pair}})}{R_A}$$

3. **Resistor between Terminals C and A:**

$$R_{CA} = R_C + R_A + \frac{R_C \cdot R_A}{R_B} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_B} = \frac{\sum(R_{\text{pair}})}{R_B}$$

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3. Step-by-Step Worked Problems

Illustration 1

Problem Statement:
Determine the current through and the power consumed by the $15\ \Omega$ resistor in the circuit shown below.

Diagram: Illustration 1 Circuit Diagram

Description of Diagram: A circuit containing multiple loops. On the far left, a $100\ \Omega$ resistor is in parallel with an independent $20\ \text{A}$ current source pointing upward. Next, a sub-network features a top $36\ \Omega$ resistor spanning across two $12\ \Omega$ series resistors, forming a central structure. Three $36\ \Omega$ vertical resistors and one central vertical $12\ \Omega$ resistor are connected. To the right, a DC voltage source of $50\ \text{V}$ in series with an internal resistance of $0.5\ \Omega$ is connected in parallel with an output branch containing a $15\ \Omega$ load resistor.

Step-by-Step Solution:

1. **Identify Node Voltage across the Target Load:**
   Observe the far right branch. The $15\ \Omega$ load resistor is directly in parallel with the $50\ \text{V}$ DC voltage source branch (which includes an internal resistance of $0.5\ \Omega$).
   However, notice the fundamental property of ideal parallel branches: if a constant independent voltage source $V_s = 50\ \text{V}$ with internal resistance $R_s = 0.5\ \Omega$ is connected across a terminal pair, the voltage delivered directly across the parallel load branch $R_L = 15\ \Omega$ is determined by the potential across those specific terminal nodes.
   Notice: The voltage directly across the rightmost branch is set by the $50\ \text{V}$ source and its $0.5\ \Omega$ resistance connected in series.
2. **Calculate Current through the $15\ \Omega$ Resistor:**
   The total equivalent resistance of the loop containing the $50\ \text{V}$ source and $15\ \Omega$ resistor (assuming source isolation/terminal evaluation):

$$R_{\text{total}} = R_{\text{internal}} + R_L = 0.5\ \Omega + 15\ \Omega = 15.5\ \Omega$$

   The current $I_{15\Omega}$ flowing through the $15\ \Omega$ resistor is:

$$I_{15\Omega} = \frac{V}{R_{\text{total}}} = \frac{50\ \text{V}}{15.5\ \Omega} \approx 3.2258\ \text{A}$$

3. **Calculate Power Consumed by the $15\ \Omega$ Resistor:**
   Using the electric power equation $P = I^2 \cdot R$:

$$P_{15\Omega} = (3.2258\ \text{A})^2 \times 15\ \Omega \approx 10.406 \times 15 \approx 156.09\ \text{W}$$

Illustration 2

Problem Statement:
Determine the power dissipated in the $10\ \Omega$ resistor.

Diagram: Illustration 2 Circuit Diagram

Description of Diagram: A bridge network connected to a DC source loop at the bottom. The DC loop contains a $12\ \mathrm{V}$ battery in series with a $10\ \Omega$ resistor. The bridge consists of outer arms with resistances $15\ \Omega$ (top left), $30\ \Omega$ (top right), $20\ \Omega$ (bottom left), and $20\ \Omega$ (bottom right). Inside the bridge, a central vertical star branch contains resistors of $6\ \Omega$ (top), $5\ \Omega$ (bottom), and a horizontal arm of $4\ \Omega$ going to the right node.

Step-by-Step Solution:

1. **Analyze the Bridge Network:**
   To find the power dissipated in the $10\ \Omega$ resistor, we need to find the total equivalent resistance (R_{eq}) of the inner bridge network attached to the voltage source loop.
2. **Identify Star (Y) Sub-Networks:**
   Focus on the inner node connecting the $6\ \Omega$, $5\ \Omega$, and $4\ \Omega$ resistors. They form a Star (Y) configuration:
   - $R_A = 6\ \Omega$
   - $R_B = 5\ \Omega$
   - $R_C = 4\ \Omega$
3. **Convert Star to Delta ($\mathrm{Y} \to \Delta$):**
   Calculate the pairwise sum product $\sum(R_{\text{pair}})$:

$$\sum(R_{\text{pair}}) = (R_A \cdot R_B) + (R_B \cdot R_C) + (R_C \cdot R_A)$$

$$\sum(R_{\text{pair}}) = (6 \times 5) + (5 \times 4) + (4 \times 6) = 30 + 20 + 24 = 74\ \Omega^2$$

   Now, find the equivalent Delta resistors (R_{AB}, R_{BC}, R_{CA}):
   - $R_{AB} = \frac{\sum(R_{\text{pair}})}{R_C} = \frac{74}{4} = 18.5\ \Omega$
   - $R_{BC} = \frac{\sum(R_{\text{pair}})}{R_A} = \frac{74}{6} = 12.333\ \Omega$
   - $R_{CA} = \frac{\sum(R_{\text{pair}})}{R_B} = \frac{74}{5} = 14.8\ \Omega$
4. **Simplify Parallel Combinations:**
   Combine the new Delta resistors with the existing outer bridge resistors that lie in parallel with them:
   - Left-top parallel pair: $R_{p1} = 15 \parallel 18.5 = \frac{15 \times 18.5}{15 + 18.5} = \frac{277.5}{33.5} \approx 8.283\ \Omega$
   - Left-bottom parallel pair: $R_{p2} = 20 \parallel 14.8 = \frac{20 \times 14.8}{20 + 14.8} = \frac{296}{34.8} \approx 8.505\ \Omega$
   - Right parallel pair: $R_{p3} = 30 \parallel 12.333 = \frac{30 \times 12.333}{30 + 12.333} = \frac{370}{42.333} \approx 8.740\ \Omega$
5. **Combine Series and Parallel Branches to Find R_{bridge}:**
   - Series combination of upper/lower branches on the left: $R_{\text{left}} = R_{p1} + R_{p2} = 8.283 + 8.505 = 16.788\ \Omega$
   - Total equivalent bridge resistance $R_{\text{bridge}} = R_{\text{left}} \parallel R_{p3} = \frac{16.788 \times 8.740}{16.788 + 8.740} = \frac{146.727}{25.528} \approx 5.747\ \Omega$
6. **Calculate Total Circuit Resistance & Current:**

$$R_{\text{total}} = R_{\text{bridge}} + R_{10\Omega} = 5.747\ \Omega + 10\ \Omega = 15.747\ \Omega$$

$$I_{\text{total}} = \frac{V}{R_{\text{total}}} = \frac{12\ \mathrm{V}}{15.747\ \Omega} \approx 0.762\ \mathrm{A}$$

7. **Calculate Power Dissipated in the $10\ \Omega$ Resistor:**

$$P_{10\Omega} = I_{\text{total}}^2 \cdot R_{10\Omega} = (0.762\ \mathrm{A})^2 \times 10\ \Omega \approx 0.5806 \times 10 \approx 5.81\ \mathrm{W}$$

Practice Problem 1

Problem Statement:
Find the equivalent resistance between terminals A and B (R_{AB}) using star-delta transformation.

Diagram: Practice 1 Circuit Diagram

Description of Diagram: A symmetrical ladder-like bridge network connected between input terminal A and output terminal B. Terminal A splits into two branches with resistors $10\ \Omega$ (upper) and $15\ \Omega$ (lower). A vertical resistor of $25\ \Omega$ connects these two branches. Following this are horizontal resistors $12\ \Omega$ (top) and $7.5\ \Omega$ (bottom), bridged by a central vertical resistor of $15\ \Omega$. Next are horizontal resistors $6.25\ \Omega$ (top) and $9.375\ \Omega$ (bottom), bridged by a vertical $25\ \Omega$ resistor. Finally, the branches converge to terminal B through resistors $5.5\ \Omega$ (upper) and $3\ \Omega$ (lower).

Step-by-Step Solution:

1. **Identify the Left Delta (Δ) Network:**
   At terminal A, the resistors $10\ \Omega$, $15\ \Omega$, and $25\ \Omega$ form a closed Delta loop.
   - $R_1 = 10\ \Omega$
   - $R_2 = 15\ \Omega$
   - $R_3 = 25\ \Omega$

   Sum of Delta resistors $\sum R_{\Delta 1} = 10 + 15 + 25 = 50\ \Omega$.
2. **Convert Left Delta to Star ($\Delta \to \mathrm{Y}$):**
   - Arm connected to Terminal A:
   $$R_{A1} = \frac{10 \times 15}{50} = \frac{150}{50} = 3\ \Omega$$
   - Top arm:
   $$R_{\text{top1}} = \frac{10 \times 25}{50} = \frac{250}{50} = 5\ \Omega$$
   - Bottom arm:
   $$R_{\text{bot1}} = \frac{15 \times 25}{50} = \frac{375}{50} = 7.5\ \Omega$$
3. **Identify the Right Delta (Δ) Network:**
   At terminal B, the resistors $5.5\ \Omega$, $3\ \Omega$, and $25\ \Omega$ form another Delta loop.
   - $R_4 = 5.5\ \Omega$
   - $R_5 = 3\ \Omega$
   - $R_6 = 25\ \Omega$

   Sum of Delta resistors $\sum R_{\Delta 2} = 5.5 + 3 + 25 = 33.5\ \Omega$.
4. **Convert Right Delta to Star ($\Delta \to \mathrm{Y}$):**
   - Arm connected to Terminal B:
   $$R_{B1} = \frac{5.5 \times 3}{33.5} = \frac{16.5}{33.5} \approx 0.493\ \Omega$$
   - Top arm:
   $$R_{\text{top2}} = \frac{5.5 \times 25}{33.5} = \frac{137.5}{33.5} \approx 4.104\ \Omega$$
   - Bottom arm:
   $$R_{\text{bot2}} = \frac{3 \times 25}{33.5} = \frac{75}{33.5} \approx 2.239\ \Omega$$
5. **Combine Simplified Series Branches:**
   - Upper horizontal series arm:
   $$R_{\text{upper\_series}} = R_{\text{top1}} + 12\ \Omega + 6.25\ \Omega + R_{\text{top2}} = 5 + 12 + 6.25 + 4.104 = 27.354\ \Omega$$
   - Lower horizontal series arm:

$$R_{\text{lower\_series}} = R_{\text{bot1}} + 7.5\ \Omega + 9.375\ \Omega + R_{\text{bot2}} = 7.5 + 7.5 + 9.375 + 2.239 = 26.614\ \Omega$$

6. **Account for Central Branch and Parallel Reductions:**
   The central vertical $15\ \Omega$ resistor connects between the midpoints of the upper and lower series paths.
   Using balanced symmetry or step-by-step reduction:
   - Total Parallel Combination $R_{\text{middle}} = R_{\text{upper\_series}} \parallel R_{\text{lower\_series}}$

$$R_{\text{middle}} = \frac{27.354 \times 26.614}{27.354 + 26.614} = \frac{728.00}{53.968} \approx 13.489\ \Omega$$

7. **Calculate Total Equivalent Resistance R_{AB}:**

$$R_{AB} = R_{A1} + R_{\text{middle}} + R_{B1}$$

$$R_{AB} = 3\ \Omega + 13.489\ \Omega + 0.493\ \Omega = 16.982\ \Omega \approx 17\ \Omega$$

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4. Key Summary Reference Table

Feature	Star Configuration (Y)	Delta Configuration (Δ)
Node Topology	3 arms meeting at 1 central common (neutral) node.	3 branches forming a closed triangular loop.
Conversion Formula	$R_A = \frac{R_{AB}R_{CA}}{\sum R_\Delta}$	$R_{AB} = \frac{\sum(R_A R_B)}{R_C}$
Denominator	Sum of all Delta resistors ($\sum R_\Delta$).	Opposite Star arm resistor (R_{opposite}).
Numerator	Product of adjacent Delta arms.	Sum of pairwise products ($\sum R_A R_B$).
Power Conservation	Total power consumed is identical in both representations.	Total power consumed is identical in both representations.